

EVOLUTION IN PROGRESS

Darwin and Wallace proposed the theory of natural selection, but it was the discovery of the gene that gave scientists the mechanism by which selection takes place. Understanding the gene has since become the key to understanding evolution.

NATURAL SELECTION

A key evolutionary mechanism, natural selection favours the survival of the fittest. In other words, individuals that possess characteristics best adapted to their current environment have a better chance of surviving to reproduce and pass on those favourable traits to another generation. Natural genetic variation within populations produces differences such as size, shape, and colour, and some of these may help promote survival. For instance, a particular body coloration may provide a camouflage that better hides the animal from predators than other colours would. If this leads to the survival of that animal and to its reproduction, that same improved coloration will be passed on to some of its offspring. As an environment



^ INDIVIDUAL VARIATION

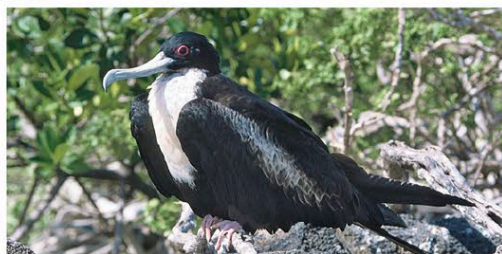
Litters of domestic cats commonly include individuals with varied coloration, especially where the parents' colours differ.

changes over time, a different coloration might be more beneficial, and so natural selection will ensure another change. A split into two populations may be imposed by a physical barrier, with each new population becoming adapted to slightly different conditions. Eventually, this may lead to one species becoming two, in a process known as speciation.



< v SEXUAL DIMORPHISM

There are often marked differences between males and females of the same species. Male frigatebirds have inflatable throat pouches for attracting the females.



SCIENCE

CREATIONISM: BELIEF VERSUS SCIENCE

Most of the world's religions provide a theory of creation, which gives an explanation for the formation of Earth and life. Many of these creation stories originated long before scientific data and theories were available to offer alternative explanations and understanding. Some believers in the Western Judeo-Christian tradition maintain that the complexity of "design" in many organisms implies that there must have been a "designer" – God – behind their creation, and for this reason they dispute the theory of evolution.

GENES AND INHERITANCE

Particular traits pass from parents to offspring through the transmission of genetic material. Genes preserve, encoded in their DNA, all the information necessary for the replication of a cell's structure and its maintenance. Genes are therefore the basic units of heredity. Individual chromosomes – the thread-like part of a cell – hold thousands of genes on long strands of DNA. During sexual reproduction, the fusion of sperm and egg cells produces two complete sets of gene-bearing chromosomes: one copy from the father and one from the mother.

